MILITARY SPECIFICATION

MANOMETER ASSEMBLIES, M159, M160,

M161, M162, AND M1

1. SCOPE

1.1 Scope. This specification covers requirements for five types of manometer assemblies.

1.2 Classification. The manometer assemblies shall be furnished in the following types as specified (see 6.2):

Type	Manometer assembly
M159	Four Tube-Vertical Manometer
M160	Three Tube-Vertical Manometer
M161	Three Tube-Vertical Manometer
M162	Two Tube-Inclined Manometer
M1	One Tube-Inclined Manometer

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-530 - Plywood, Flat-Panel.
QQ-S-781 - Steel, Strapping, Flat.
TT-W-00572 - Wood-Preservative; Water-Repellent.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-P-291 - Paperboard, Wrapping, Cushioning.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-2427 - Boxes, Ammunition Packing; Wood Nailed.
MIL-C-7769 - Cushioning Material, Uncompressed Bound Fiber for Packaging.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

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DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL5-19-5396	-	Manometer Assembly, 600 CFM Filter Unit, Four Tube, M159.
DL5-19-5397	-	Manometer Assembly, 600 CFM Filter Unit, Three Tube, M160.
DL5-19-5398	-	Manometer Assembly, 300 CFM Filter Unit, Three Tube, M161.
DL5-19-5399	-	Manometer Assembly, Protective Shelter, Two Tube, M162.
DL5-19-5400	-	Manometer Assembly, Protective Shelter, One Tube, M1.
D150-2-1	-	Marking Diagram for Shipping Containers.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules, and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on the applicable data list for the respective manometer assembly shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of their respective manometer assemblies shall conform to the specifications and drawings listed on the applicable data list.

3.2 Assembly. The manometer assemblies shall be assembled in accordance with the following applicable drawing:

Type	Drawing
M159	D5-19-5396
M160	D5-19-5397
M161	D5-19-5398
M162	D5-19-5399
M1	D5-19-5400

3.3 Leakage. The empty manometer assembly shall be capable of maintaining 14 $\pm$ 1.0 inch water pressure for 5 $\pm$ 0.5 minute when tested as specified in 4.4.4.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of manometer assemblies shall be produced in accordance with this specification for examination and test (see 4.3).

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3.5 Workmanship. The manometer assembly shall be free from damage (crimped, cracked, or broken tubing; cracked or chipped manometer tube; cracked or broken tube nuts; and cracked or broken mounting panel) and contamination (dirt, oil, grease, or foreign matter).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 2 manometer assemblies of each type specified in the contract or order shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure for examination and nondestructive test. The preproduction sample of manometer assemblies shall be examined in accordance with the classification of defects 4.4.3.2 and tested in accordance with 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the manometer assemblies of one type, produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedures.

4.4.3.1 For examination and nondestructive test. Sample manometer assemblies and the level A and B preparation for delivery shall be examined and tested in accordance with the classification of defects (4.4.3.2) and MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Monometer assembly, M159, M160, M161, M162, and M1 (Dwg D5-19-5396, D5-19-5397, D5-19-5398, D5-19-5399, and D5-19-5400).

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Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Components missing or incorrectly assembled	
102	Manometers not as specified	
103	Marking of designation label and instruction card	
104	Workmanship (see 3.5)	
105	Three-way vent valve not free to open and close by hand	
106	Leakage	4.4.4

4.4.3.2 Classification of defects (Continued).

(b) Preparation for delivery (see section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Packing method not as specified
102	Outer container not as specified
103	Cushioning material missing or not as specified
104	Boxes not coated with wood preservative, when applicable
105	Required number of ampoules of manometer fluid for each manometer assembly missing, incorrect, or inadequate
106	Marking

4.4.4 Leakage test. Test shall be conducted as follows: Connect an aspirator bulb and standard water manometer in parallel with the manometer assembly under test. Install tubing and clamps as necessary to interconnect the individual manometers of the manometer assembly, making certain all such connections are tight. Apply pressure by means of the aspirator bulb until the specified reading is obtained on the standard water manometer. Close off the air supply by means of a pinch clamp or valve and hold the pressure for the required period of time. A decrease in pressure is an indication of leakage.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Level A.

5.1.1 Red indicating liquid, specific gravity 0.834, 3/4 ounce. The required number of ampoules of manometer fluid for each manometer assembly, as specified on applicable Drawings D5-19-5396, D5-19-5397, D5-19-5398, D5-19-5399, and D5-19-5400, shall be wrapped in paperboard conforming to type I, style 1 of PPP-P-291 and secured with pressure sensitive tape conforming to PPP-T-76.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each manometer assembly, together with items packaged as specified in 5.1, shall be packaged in accordance with the requirements of submethod IA-14 of MIL-P-116. The outer container shall be either a nailed wood box conforming to class 2, overseas, style 2 of PPP-B-621 or an overseas type, style A, cleated plywood box conforming to PPP-B-601 and using plywood complying with group A of NN-P-530 in accordance with type I, grade 3 or 4 of CS-35 provided the plywood does not have continuous holes or openings extending through the entire panel from face to back. The sides, ends, bottom and top of the box shall be fitted with nominal two inch thick cushioning material conforming to type II of MIL-C-7769. The wooden box shall be closed and strapped in accordance with the appendix to the box specification. Strapping conforming to type I, class B of QQ-S-781 shall be used.

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5.2.1.1 Wood preservative. The wood boxes or the unpainted finished wood parts thereof, specified in 5.2.1, shall be preserved by completely immersing for a minimum of three minutes in a water repellent wood preservative conforming to composition A of TT-W-00572 and then dried. Prior to packing the items in the box, the treated wooden box shall meet the dryness of preservative requirement specified in MIL-B-2427.

5.2.2 Level B. Each manometer assembly, together with items packaged as specified in 5.1, shall be packaged in accordance with the requirements of submethod IA-14 of MIL-P-116. The sides, ends, bottom and top of the box shall be fitted with nominal two inch thick cushioning material conforming to type II of MIL-C-7769.

5.2.3 Level C. The manometer assemblies, together with items packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1. All marking on the shipping containers shall be clear, legible, and permanent after the application of the wood preservative specified in 5.2.1.1.

5.4 Repair parts. When components of this item are procured for storage and issue as a repair part, preservation, packaging, packing, and marking shall be as specified on the packaging data sheet which is identified by the Federal Stock Number.

6. NOTES

6.1 Intended use. The manometer assemblies covered by this specification are for use in Collective Protection Equipment, CBR; Nike-Hercules CONUS: M11, M12 and M13.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of manometer assembly required (see 1.2).
(c) Level of packing required (see 5.2).
(d) Preproduction:

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

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